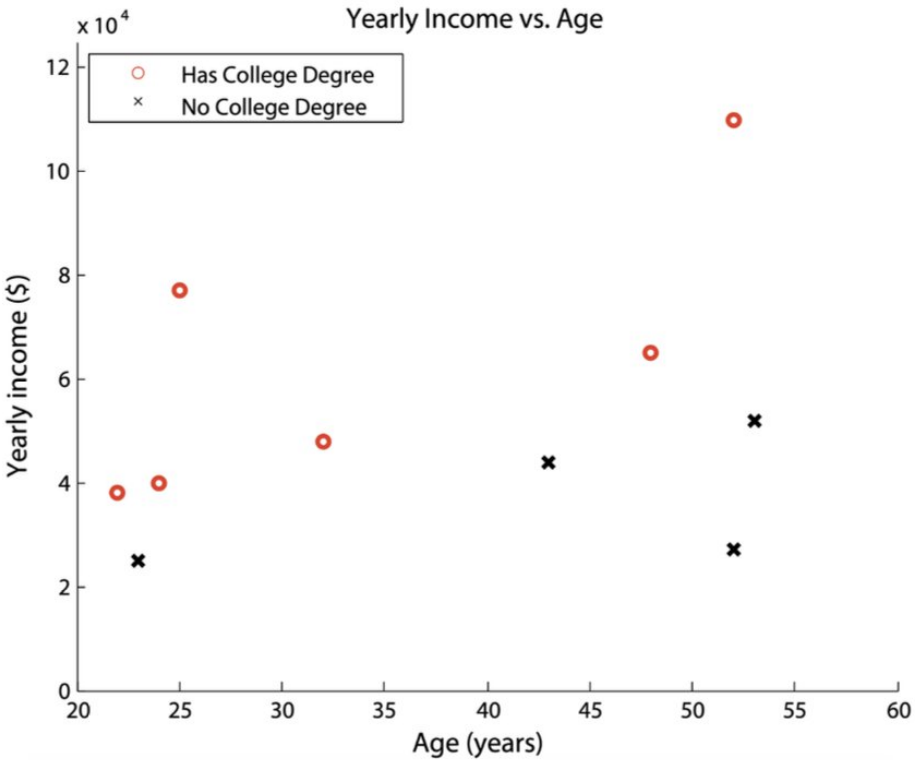
1. Consider the problem of predicting if a person has a college degree based on age and salary. The table and graph below contain training data for 10 individuals.

Age	Salary	College Degree
24	40,000	Yes
53	52,000	No
23	24,000	No
25	77,000	Yes
32	48,000	Yes
52	108,000	Yes
22	38,000	Yes
43	44,000	No
52	27,000	No
48	67,000	Yes



a.

原始数据的信息熵

$$H = -\frac{2}{5}\log\frac{2}{5} - \frac{3}{5}\log\frac{3}{5} \approx 0.97$$

以年龄（age=37.5）为分割点时

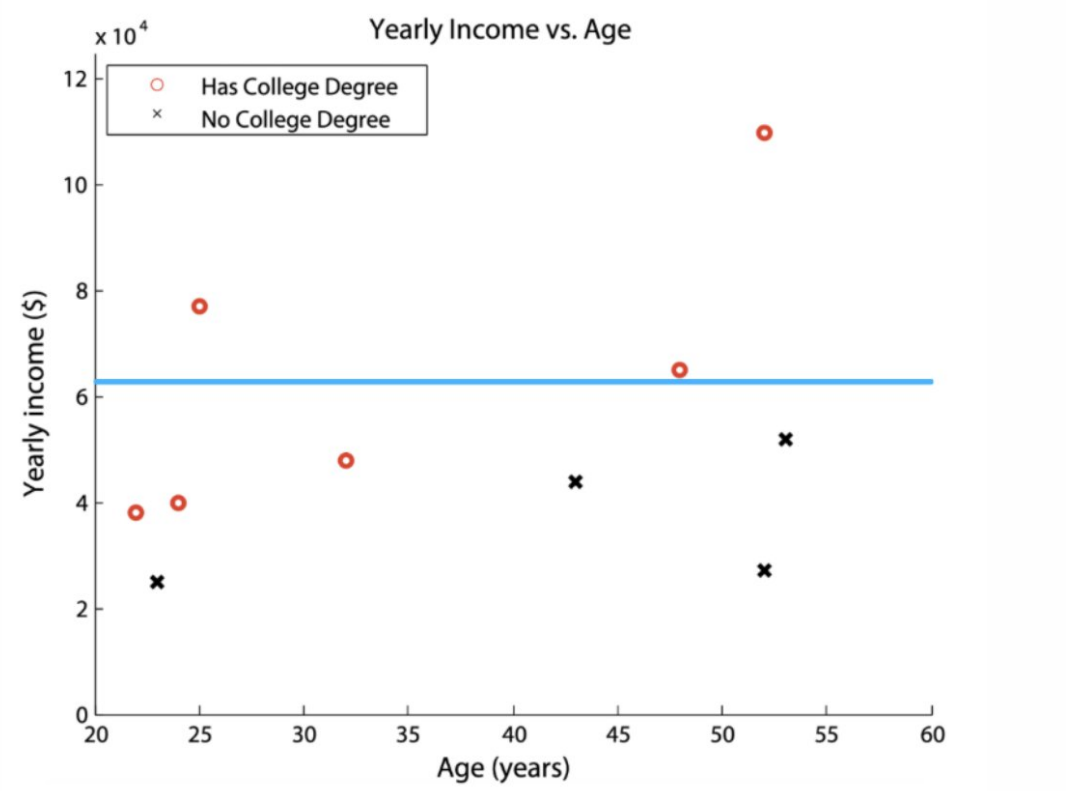
$$H(D \mid Age) = \left(-\frac{1}{5}\log\frac{1}{5} - \frac{4}{5}\log\frac{4}{5}\right) \times \frac{1}{2} + \left(-\frac{2}{5}\log\frac{3}{5} - \frac{2}{5}\log\frac{2}{5}\right) \times \frac{1}{2} = 0.85$$

以薪资（salary=66,000）为分割点时

$$H(D \mid Salary) = \left(-\frac{3}{7}\log\frac{3}{7} - \frac{4}{7}\log\frac{4}{7}\right) \times \frac{7}{10} + \left(-\frac{3}{3}\log\frac{3}{3} - \frac{0}{3}\log\frac{0}{3}\right) \times \frac{3}{10} = 0.69$$

因此，**使用薪资作为分割点可以获得最大的最大信息增益**

$$IG = 0.97 - 0.69 = 0.28$$



原始数据集的信息熵

$$H(D) = -\frac{3}{7}\log\frac{3}{7} - \frac{4}{7}\log\frac{4}{7} = 0.99$$

以年龄（age=37.5）为分割点时

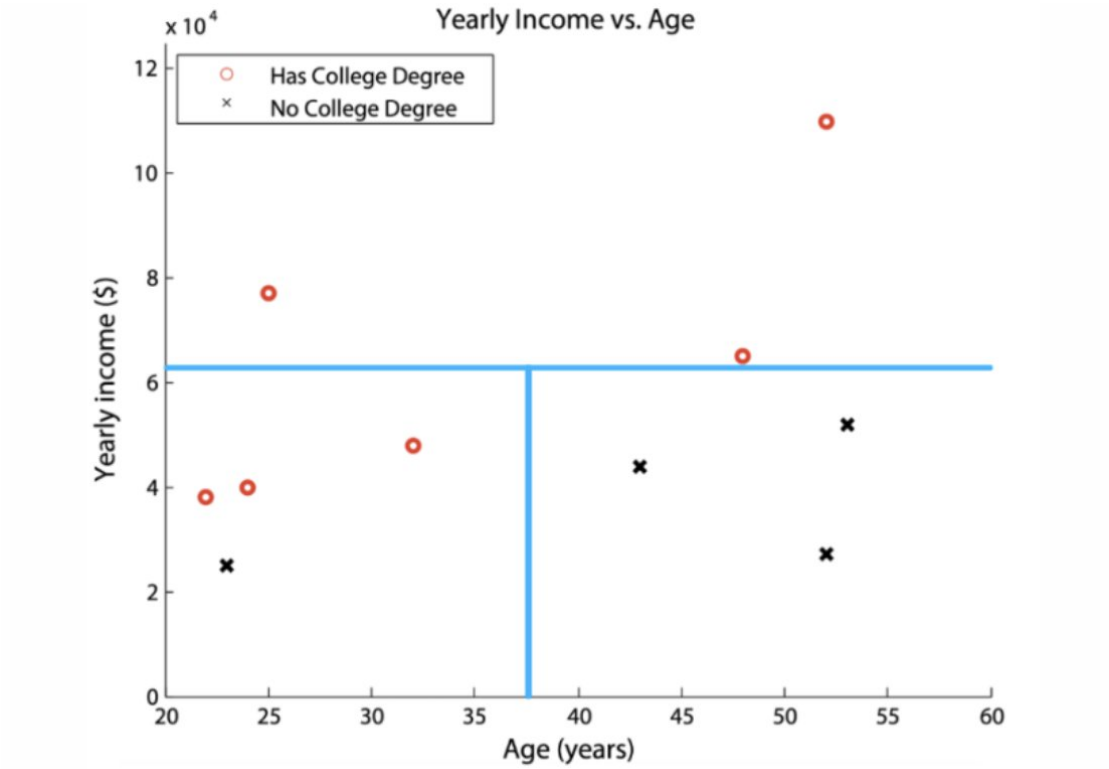
$$H(D \mid Age) = \frac{3}{7} \times 0 + \frac{4}{7} \times \left(-\frac{1}{4}\log\frac{1}{4} - \frac{3}{4}\log\frac{3}{4}\right) = 0.46$$

以薪资（salary=38,000）为分割点时

$$H(D \mid Salary) = \frac{5}{7} \times \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) + \frac{2}{7} \times 0 = 0.69$$

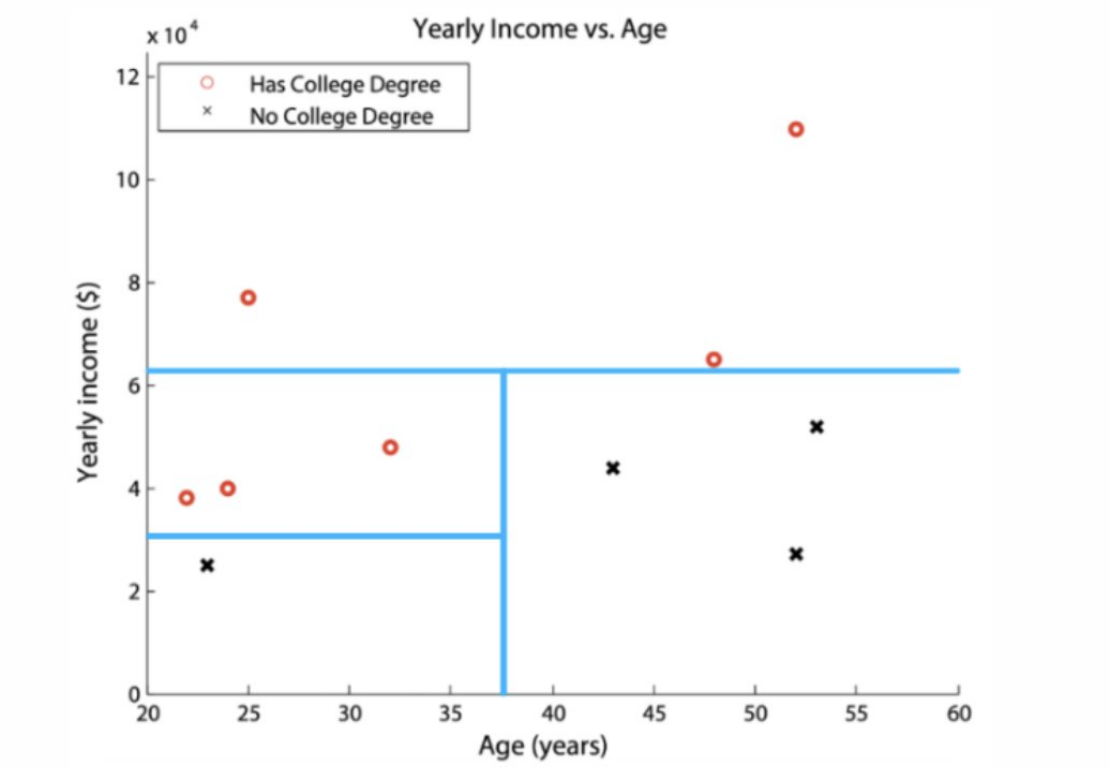
所以我们选择年龄 age=37.5 作为分割点，此时的信息增益为：

$$IG = 0.99 - 0.46 = 0.53$$

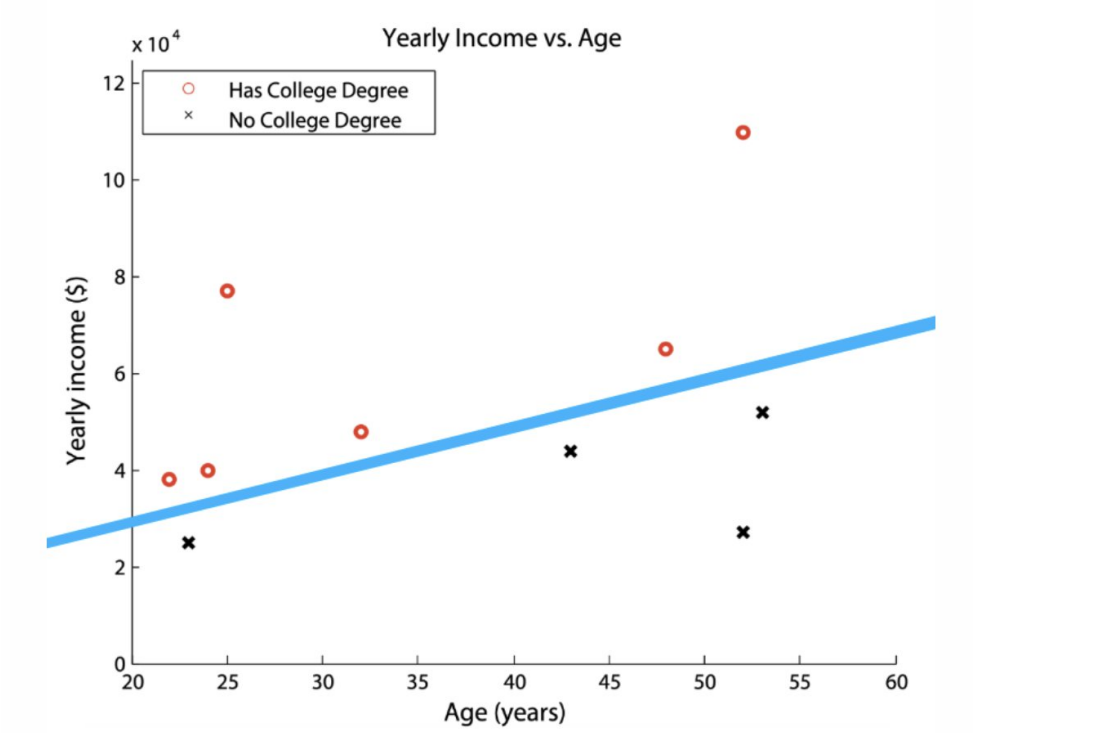


最后一次分裂，显然我们选择工资等于36000作为分裂条件

$$IG = \left(-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right) - 0 = 0.81$$



b.



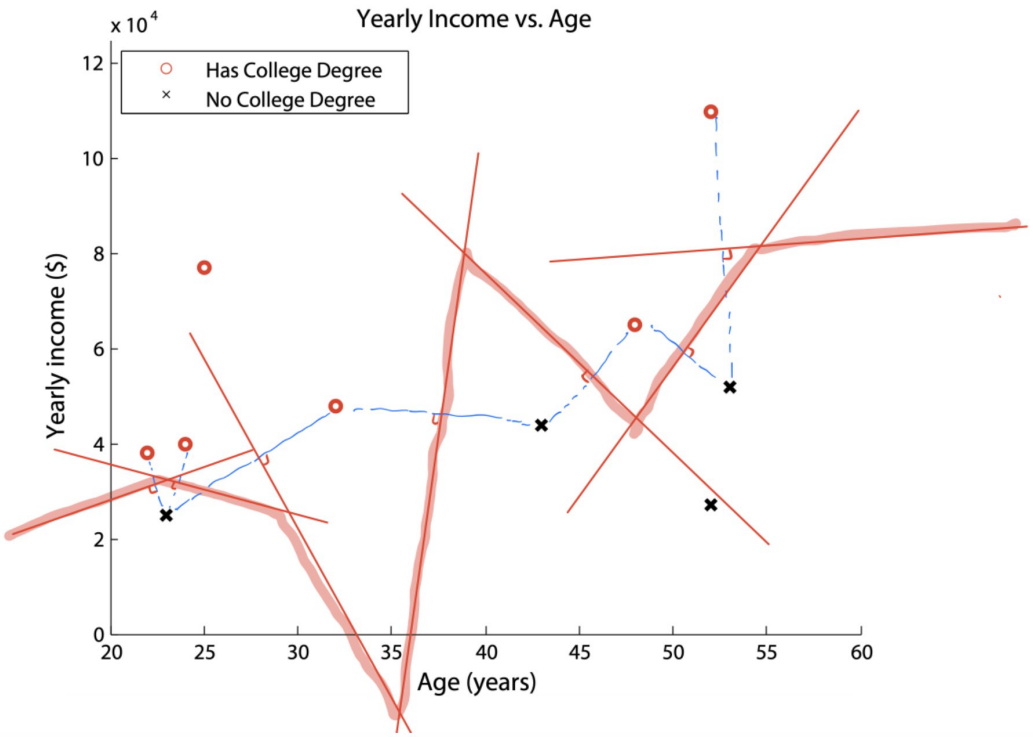
只需要选定具体的参数，使得线满足把两类完全分开

例如α=-0.1，β=-0.0001

那么信息增益计算为

$$IG = 0.97 - 0 = 0.97.$$

c.



2.

记

训练集文档数 $m = 6$，类别集合 $C = \{\text{Like}, \text{Dislike}\}$

分词后词汇表大小 $|V| = 27$

记

N_c — 类别 c 的总词频

$N_{c,w}$ — 词 w 在类别 c 中出现次数

先验概率计算为：

$$P(c) = \frac{\#\text{文档属 } c}{m}, \qquad P(\text{Like}) = P(\text{Dislike}) = \frac{3}{2} \Big/ 6 = \frac{1}{2}$$

带 Laplace 平滑的条件概率

$$P(w \mid c) = \frac{N_{c,w} + 1}{N_c + |V|}$$

词 w	$N_{\text{Like},w}$	$N_{\text{Dis},w}$	$P(w \mid \text{Like})$	$P(w \mid \text{Dis})$
and	1	0	$\frac{1+1}{47} = 0.04255$	$\frac{0+1}{45} = 0.02222$
cheap	1	1	0.04255	0.04444
delicious	2	1	0.06383	0.04444
very	1	0	0.04255	0.02222

w=[very, delicious, and, cheap] 的似然

$$P(\mathbf{w} \mid \text{Like}) = \prod_{w \in \mathbf{w}} P(w \mid \text{Like}) = 4.918 \times 10^{-6},$$
$$P(\mathbf{w} \mid \text{Dislike}) = 9.755 \times 10^{-7}.$$

后验概率

$$\tilde{P}(c \mid \mathbf{w}) = P(c)\, P(\mathbf{w} \mid c), \qquad P(c \mid \mathbf{w}) = \frac{\tilde{P}(c \mid \mathbf{w})}{\sum_{c'} \tilde{P}(c' \mid \mathbf{w})}$$

$$P(\text{Like} \mid \mathbf{w}) = 0.83449,$$
$$P(\text{Dislike} \mid \mathbf{w}) = 0.16551.$$

所以答案是Like